

# Week 9 Lab 3: Nozzle Data Alignment Audit

FlowMLLab – AI in Fluid Mechanics

## Purpose and scope

Extend the Week 4 branch–trunk data contract to moving-throat nozzle grids. This CPU exercise uses an ideal-gas, inviscid, isentropic, choked, shock-free quasi-one-dimensional reference. It does not train a neural model or validate multidimensional CFD. The exercise is independently written.

## Three objects, one physical pairing

For nozzle  $c$  and query  $j$ ,

$$B_c = [A_c(\xi_1)/A_t, \dots, A_c(\xi_{41})/A_t, p_{0,c}, T_{0,c}], \quad (B_c, x_{cj}) \mapsto q_c(x_{cj}).$$

The sensor locations  $\xi_i$  are fixed. The geometry values vary. Query grids may differ; here each includes its own throat exactly. Targets are  $q = (M, p, T, \rho, u)$ . The branch and target case IDs must agree, and the query must equal the source target coordinate in the same units.

## Intentional failures

The notebook first repeats the first nozzle’s query grid for every nozzle. All array shapes remain correct, but the coordinate audit rejects the pairing. It then incorrectly uses the case-dependent query grids as fixed branch sensors; the sensor audit rejects this too. These are expected caught errors, not failed notebook execution.

## Repair and reference checks

Keep each target’s own query location; evaluate branch geometry at the common 41 sensors. Alternatively, evaluate all reference fields directly at common queries. If CFD fields are interpolated instead, measure interpolation error separately. Check positivity,  $|M(x_t) - 1|$ , area–Mach consistency and

$$C_m = \frac{\text{std}(\rho u A / A_t)}{\text{mean}(\rho u A / A_t)}.$$

These checks verify the reference. Passing them does not demonstrate neural accuracy. An audit also cannot discover metadata that was already mislabeled by the exporter; retain independent source coordinates.

## From debugging to a model comparison

Split complete physical cases before expanding spatial rows. Fit normalization and any POD basis on selection-training cases only. Validation supports model selection and early stopping. Freeze MLP and DeepONet before a fresh final test; previously inspected cases are diagnostic data. Report per-case relative  $L_2$  mean, standard deviation and maximum; physical violations; and warmed-up full-profile prediction time. A new random seed is not proof of disjoint cases. The two-parameter geometry family is not evidence of arbitrary-shape generalization.

## Questions

Why can loss decrease on incorrectly paired data? Why does repeating a branch row differ from tiling the branch matrix? Why need sensors and queries not coincide? What evidence distinguishes metadata consistency, reference validity, and neural accuracy? Explain before adding a more complex architecture.